

Nonlinear Robust Control for Dual-UAV Collaborative Aerial Transportation System

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Abstract—Quadrotor UAVs have the advantages of high flexibility and mobility, and the use of UAVs for transportation tasks can avoid the restrictions of complex terrain while improving transportation efficiency and reducing transportation costs. However, as the load capacity of a single UAV is very limited, multiple UAVs are introduced for coordinated transport to increase the system's capacity. Aiming at transporting bar-shaped cargos, this paper studies the dual-UAV cooperative transportation system. However, compared with the single UAV transportation system, the dual-UAV cooperative transportation system has higher degrees of freedom, stronger nonlinearity, and more complex state coupling. Besides, considering the uncertainty and complexity of the actual environment, the system may encounter external disturbances during actual flight. Thus, to overcome the aforementioned challenges, an effective control scheme is urgently needed to ensure precise UAV positioning and cargo swing suppression for smooth cargo transportation. Therefore, this paper proposes a nonlinear robust control scheme based on the finite-time disturbance observer for the dual-UAV collaborative aerial transportation system. Subsequently, the convergence of the equilibrium point is proved by Lyapunov techniques and LaSalle's Invariance Theorem. Finally, the excellent performance and robustness of the proposed control scheme are verified by groups of simulations.

Index Terms—Collaborative aerial transportation system, nonlinear control, finite-time disturbance observer

I. INTRODUCTION

With the continuous improvement of artificial intelligence technology, the control technologies of quadrotor unmanned aerial vehicles (UAVs) have come to maturity, which are widely used in various fields such as logistic transportation, agricultural plant protection, security and rescue, geographic mapping, and webcasting [1]–[3]. Specifically, UAVs play an important role in the field of aerial transportation due to their high flexibility, and maneuverability.

Currently, the most widely used method of the aerial transportation system is to suspend the cargo beneath the UAV through cable [4]–[6], which is cost-effective and has no limitation on the cargo size. Based on the backstepping technique,

a nonlinear control scheme [7] is designed for load tracking a predetermined trajectory. For UAV positioning and cargo swing elimination, an enhanced coupling hierarchical control law is presented in [8], and the asymptotic stability of the closed-loop system is guaranteed with Lyapunov techniques and LaSalle's Invariance Theorem. A reinforcement learning network is designed to parameterize the nonlinear controller to obtain an algorithm with high reliability and fast convergence in [9], which has the ability to deal with complex operating conditions such as model uncertainties, parameters drift, and external disturbances.

However, the load capacity of a single UAV is very limited, it is an urgent need to introduce multiple UAVs for cooperative aerial transportation. In [10], the UAVs and the load are treated as single systems, and the kinematics is described as a multi-robot formation utilizing the virtual structure approach. Then, an adaptive dynamic compensator is proposed to accomplish load positioning, orientation, and trajectory tracking under adverse conditions. [11] studies a bar tethered to two UAVs, aiming to maintain the bar stable around the desired pose. The equations of motion of the system are derived, PID control laws are implemented for each UAV to achieve closed-loop control and the closed-loop vector field is linearized around some equilibrium points of interest. A UAV-load unified signal [12] is constructed to enhance the state coupling between the UAV positions and cargo swing angles, which greatly improves the swing elimination performance.

Compared to a single quadrotor, the collaborative aerial transportation system presents much more complex underactuated characteristics, while cargo swing enhances the dynamic coupling and nonlinearity of the system, making the control problem more difficult. Moreover, in practical applications, the system faces many uncertainties, such as wind disturbances, uncertainty of system parameters, etc. Therefore, there is an urgent need to design a nonlinear robust control method for multi-UAV collaborative aerial transportation systems.

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To address the above issues, this paper designs a robust control scheme for the collaborative aerial transportation system based on the finite time disturbance observer. Specifically, to overcome the disturbance, a disturbance observer is first designed, and then the controller is designed by energy-based analysis. Subsequently, the stability of the closed-loop system is proved using Lyapunov techniques and LaSalle's Invariance Theorem. Finally, the performance of the proposed controller is verified by simulation. The proposed control scheme makes some significant contributions. Firstly, the proposed observer can achieve an accurate and unbiased estimation of the external disturbances of the system in finite time theoretically, thus significantly mitigating the negative effects of the disturbances. Secondly, the proposed control scheme ensures accurate and fast positioning of the quadrotor UAVs while suppressing cargo swing and is robust to external disturbances.

The rest of this paper is organized as follows. Section II gives the dynamic model of the collaborative aerial transportation system and describes the control objective. Section III provides the details of disturbance estimation, controller design, and stability analysis. Then, in Section IV, the performance and robustness of the proposed control scheme are verified by simulations. Finally, Section V is a summary of this paper.

Notation: Throughout this paper, the following abbreviated expressions are defined: $C_i = \cos \theta_i$, $C_{i\pm j} = \cos(\theta_i \pm \theta_j)$, $S_i = \sin \theta_i$, $S_{i\pm j} = \sin(\theta_i \pm \theta_j)$, $i, j = 1, 2, 3$. For a vector $\mathbf{h} \in \mathbb{R}^n$, define $\mathbf{h}^{[p]} = [\text{sgn}(h_1)|h_1|^p, \dots, \text{sgn}(h_n)|h_n|^p]^\top$, $\text{sgn}(\mathbf{h}) = [\text{sgn}(h_1), \dots, \text{sgn}(h_n)]^\top$, $|\mathbf{h}| = [|h_1|, \dots, |h_n|]^\top$, $\cosh(\mathbf{h}) = [\cosh(h_1), \dots, \cosh(h_n)]^\top$, $\ln(\mathbf{h}) = [\ln(h_1), \dots, \ln(h_n)]^\top$, $\tanh(\mathbf{h}) = [\tanh(h_1), \dots, \tanh(h_n)]^\top$, and $\text{diag}(\mathbf{h})$ is a diagonal matrix with diagonal elements of $h_1, \dots, h_n$.

II. PROBLEM FORMULATION

A. Dynamic Modeling

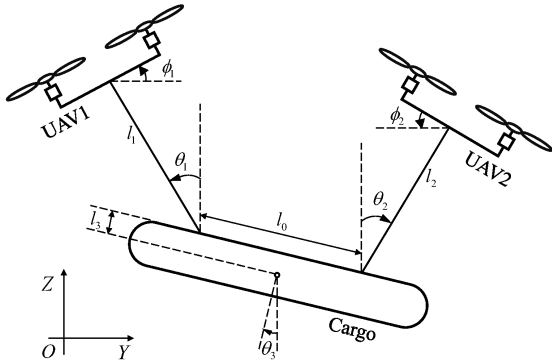


Fig. 1. Illustration of the collaborative aerial transportation system.

Fig. 1 is the illustration of the collaborative aerial transportation system with a suspended bar-shaped cargo connected by dual quadrotor UAVs. The coordinate system Y - O - Z is an inertial reference frame fixed to the earth. $M_1, M_2, m, l_1, l_2, l_3 \in \mathbb{R}$ denote the quadrotor masses, cargo mass, cable lengths, and half of the cargo thickness. $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^\top \in \mathbb{R}^3$, where θ_1, θ_2 denote angles between the two cables and the

positive half-axis of the Z -axis, θ_3 denotes deflection angle of the cargo, $\boldsymbol{\xi}_i = [y_i, z_i]^\top \in \mathbb{R}^2$, $i = 1, 2$ denotes the position of each quadrotor's mass center in the inertial reference frame.

Based on Euler-Lagrange modeling methods, the dynamic equation describing the whole system is provided as follows:

$$M_c(\mathbf{q})\ddot{\mathbf{q}} + V_c(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{U} + \mathbf{D}, \quad (1)$$

wherein $\mathbf{q} = [y_1, z_1, y_2, z_2, \theta_1, \theta_2, \theta_3]^\top \in \mathbb{R}^7$ denotes the state vector of quadrotor UAVs position and cargo attitude, $M_c(\mathbf{q}), V_c(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{7 \times 7}$, and $\mathbf{G}(\mathbf{q}), \mathbf{U}, \mathbf{D} \in \mathbb{R}^7$ denote the inertia matrix, centripetal-Coriolis matrix, gravity vector, control vector, and external disturbance vector, taking the following form as $\mathbf{G} = [0, 0, 0, 0, \frac{1}{2}mgl_1S_1, \frac{1}{2}mgl_2S_2, mgl_3S_3]^\top$, $\mathbf{U} = [u_1, u_2, u_3, u_4, 0, 0, 0]^\top$, $\mathbf{D} = [d_1, d_2, d_3, d_4, 0, 0, 0]^\top$, wherein u_1, u_2, u_3, u_4 denote the component forces of the two UAVs on the axes Y and Z respectively, whose detailed expressions are given as $u_1 = f_1 \sin \phi_1$, $u_2 = f_1 \cos \phi_1 - (M_1 + \frac{1}{2}m)g$, $u_3 = f_2 \sin \phi_2$, $u_4 = f_2 \cos \phi_2 - (M_2 + \frac{1}{2}m)g$, wherein f_1 and f_2 denote the thrusts of the UAVs, ϕ_1 and ϕ_2 denote the attitude angles of the UAVs, g denotes the gravity coefficient.

To facilitate controller development and stability analysis, (1) is expanded to the following form:

$$M_1\ddot{y}_1 + \frac{1}{4}m(\ddot{y}_1 + \ddot{y}_2 + l_1C_1\ddot{\theta}_1 - l_2C_2\ddot{\theta}_2 + 2l_3C_3\ddot{\theta}_3 - l_1S_1\dot{\theta}_1^2 + l_2S_2\dot{\theta}_2^2 - 2l_3S_3\dot{\theta}_3^2) = u_1 + d_1, \quad (2)$$

$$M_1\ddot{z}_1 + \frac{1}{4}m(\ddot{z}_1 + \ddot{z}_2 + l_1S_1\ddot{\theta}_1 + l_2S_2\ddot{\theta}_2 + 2l_3S_3\ddot{\theta}_3 + l_1C_1\dot{\theta}_1^2 + l_2C_2\dot{\theta}_2^2 + 2l_3C_3\dot{\theta}_3^2) = u_2 + d_2, \quad (3)$$

$$M_2\ddot{y}_2 + \frac{1}{4}m(\ddot{y}_1 + \ddot{y}_2 + l_1C_1\ddot{\theta}_1 - l_2C_2\ddot{\theta}_2 + 2l_3C_3\ddot{\theta}_3 - l_1S_1\dot{\theta}_1^2 + l_2S_2\dot{\theta}_2^2 - 2l_3S_3\dot{\theta}_3^2) = u_3 + d_3, \quad (4)$$

$$M_2\ddot{z}_2 + \frac{1}{4}m(\ddot{z}_1 + \ddot{z}_2 + l_1S_1\ddot{\theta}_1 + l_2S_2\ddot{\theta}_2 + 2l_3S_3\ddot{\theta}_3 + l_1C_1\dot{\theta}_1^2 + l_2C_2\dot{\theta}_2^2 + 2l_3C_3\dot{\theta}_3^2) = u_4 + d_4, \quad (5)$$

$$\frac{1}{4}ml_1(C_1\ddot{y}_1 + S_1\ddot{z}_1 + C_1\ddot{y}_2 + S_1\ddot{z}_2 + l_1\ddot{\theta}_1 - l_2C_{1+2}\ddot{\theta}_2 + 2l_3C_{1-3}\ddot{\theta}_3 + l_2S_{1+2}\dot{\theta}_2^2 + 2l_3S_{1-3}\dot{\theta}_3^2) + \frac{1}{2}mgl_1S_1 = 0, \quad (6)$$

$$\frac{1}{4}ml_2(-C_2\ddot{y}_1 + S_2\ddot{z}_1 - C_2\ddot{y}_2 + S_2\ddot{z}_2 - l_1C_{1+2}\ddot{\theta}_1 + l_2\ddot{\theta}_2 - 2l_3C_{2+3}\ddot{\theta}_3 + l_1S_{1+2}\dot{\theta}_1^2 + 2l_3S_{2+3}\dot{\theta}_3^2) + \frac{1}{2}mgl_2S_2 = 0, \quad (7)$$

$$\frac{1}{2}ml_3(C_3\ddot{y}_1 + S_3\ddot{z}_1 + C_3\ddot{y}_2 + S_3\ddot{z}_2 + l_1C_{1-3}\ddot{\theta}_1 - l_2C_{2+3}\ddot{\theta}_2 + 2l_3\ddot{\theta}_3 - l_1S_{1-3}\dot{\theta}_1^2 + l_2S_{2+3}\dot{\theta}_2^2) + mgl_3S_3 = 0. \quad (8)$$

Based on the actual situation, the following assumptions can be made [13]–[16]:

Assumption 1: Considering the actual working conditions of the cooperative aerial transportation system. The cable and

cargo swing angle satisfy:

$$-\frac{\pi}{2} < \theta_i(t) < \frac{\pi}{2}, i = 1, 2, 3, \forall t \geq 0.$$

Assumption 2: The derivative of the disturbance $\dot{d}_i$ is bounded and satisfies $|\dot{d}_i| \leq F_i$, wherein F_i is the constant, for $i = 1, 2, 3, 4$.

Property 1: The inertia matrix $M_c(q)$ is a positive definite symmetric matrix. Moreover, the centripetal-Coriolis matrix $V_c(q, \dot{q})$ and the time derivative of $M_c(q)$ satisfy the following skew-symmetric relationship:

$$\mathbf{x}^\top \left(\frac{1}{2} \dot{M}_c - V_c \right) \mathbf{x} = 0, \forall \mathbf{x} \in \mathbb{R}^7.$$

B. Control Objective

The control objective of the cooperative aerial transportation system is to suppress the cargo swing as much as possible while driving the quadrotors from the initial position to their desired position $\xi_{id} = [y_{id}, z_{id}]^\top, i = 1, 2$ accurately and quickly in the presence of external disturbances. It can be mathematically described as

$$\xi_1 \rightarrow \xi_{1d}, \xi_2 \rightarrow \xi_{2d}, \theta \rightarrow \theta_{3 \times 1}. \quad (9)$$

The quadrotors position error is defined as $e_{\xi_i}(t) = \xi_i(t) - \xi_{id}(t), i = 1, 2$, thus, the state error can be defined as $e_q = [e_{\xi_1}^\top, e_{\xi_2}^\top, \theta^\top]^\top$.

III. DISTURBANCE PROCESSING AND CONTROLLER DESIGN

A. Disturbance Observer

To facilitate the subsequent analysis, one can elaborately construct a virtual signal χ for the system, which is given as

$$\chi = \begin{bmatrix} (M_1 + \frac{1}{4}m)y_1 + \frac{1}{4}my_2 + \frac{1}{4}m(l_1S_1 - l_2S_2 + 2l_3S_3) \\ (M_1 + \frac{1}{4}m)z_1 + \frac{1}{4}mz_2 + \frac{1}{4}m(-l_1C_1 - l_2C_2 - 2l_3C_3) \\ \frac{1}{4}my_1 + (M_1 + \frac{1}{4}m)y_2 + \frac{1}{4}m(l_1S_1 - l_2S_2 + 2l_3S_3) \\ \frac{1}{4}mz_1 + (M_1 + \frac{1}{4}m)z_2 + \frac{1}{4}m(-l_1C_1 - l_2C_2 - 2l_3C_3) \end{bmatrix},$$

thus, model (2)-(5) can be transformed into the following compact form:

$$\ddot{\chi} = U_\chi + D_\chi.$$

wherein $U_\chi = [u_1, u_2, u_3, u_4]^\top, D_\chi = [d_1, d_2, d_3, d_4]^\top$.

Subsequently, define an auxiliary variable $e_\chi = \dot{\chi} - \delta_\chi$, and the intermediate variable δ_χ takes the following form:

$$\dot{\delta}_\chi = U_\chi + c_{\alpha_\chi} e_\chi^{[\alpha_\chi]} + c_{\beta_\chi} e_\chi^{[\beta_\chi]} + \xi_\chi,$$

wherein the coefficients matrices $c_{\alpha_\chi}, c_{\beta_\chi} \in \mathbb{R}_+^{4 \times 4}$ are positive definite diagonal matrices, $0 < \alpha_\chi < 1, \beta_\chi > 1$ are positive parameters, ξ_χ is intermediate variables to-be-constructed.

Subsequently, define the sliding surface as

$$s_\chi = \dot{e}_\chi + c_{\alpha_\chi} e_\chi^{[\alpha_\chi]} + c_{\beta_\chi} e_\chi^{[\beta_\chi]}, \quad (10)$$

and the disturbance observer of D_χ is designed as

$$\dot{\hat{D}}_\chi = \xi_\chi, \quad (11)$$

$$\dot{\xi}_\chi = (\lambda_{d_\chi} + \mu_{d_\chi}) \text{sgn}(s_\chi), \xi_\chi(0) = \mathbf{0}_{4 \times 1}, \quad (12)$$

wherein $\lambda_{d_\chi}, \mu_{d_\chi} \in \mathbb{R}_+^{4 \times 4}$ are positive definite diagonal matrices. The specific forms are $\lambda_{d_\chi} = \text{diag}([\lambda_{d_1}, \lambda_{d_2}, \lambda_{d_3}, \lambda_{d_4}])$, $\mu_{d_\chi} = \text{diag}([\mu_{d_1}, \mu_{d_2}, \mu_{d_3}, \mu_{d_4}])$, wherein μ_{d_i} satisfies $\mu_{d_i} > F_i$, for $i = 1, 2, 3, 4$.

Lemma 1: Consider the following scalar system [17], [18]:

$$\dot{\nu} = -c_\alpha \text{sgn}(\nu)|\nu|^\alpha - c_\beta \text{sgn}(\nu)|\nu|^\beta, \quad \nu(0) = \nu_0,$$

wherein c_α, c_β are positive constants, $0 < \alpha < 1$ and $\beta > 1$. The equilibrium point of the above system is finite time stable and its settling time is bounded by: $T < T_{\max} = \frac{1}{c_\alpha} \frac{1}{1-\alpha} + \frac{1}{c_\beta} \frac{1}{1-\beta}$.

Theorem 1: The disturbance observer (11)-(12) guarantees that the disturbance estimation error of D_χ converges to zero in finite time, under *Assumption 2* and the conditions $0 < \alpha_\chi < 1, \beta_\chi > 1, \mu_{d_i} > F_i$ for $i = 1, 2, 3, 4$.

Proof 1: The time derivative of e_χ with respect to time is derived as follows:

$$\dot{e}_\chi = \ddot{\chi} - \dot{\delta}_\chi = D_\chi - c_{\alpha_\chi} e_\chi^{[\alpha_\chi]} - c_{\beta_\chi} e_\chi^{[\beta_\chi]} - \xi_\chi. \quad (13)$$

Then, comparing (13) with (10), one can obtain that $s_\chi = D_\chi - \xi_\chi$. Substituting (12) into the time derivative of the sliding surface s_χ leads to

$$\dot{s}_\chi = \dot{D}_\chi - \dot{\xi}_\chi = \dot{D}_\chi - (\lambda_{d_\chi} + \mu_{d_\chi}) \text{sgn}(s_\chi). \quad (14)$$

Subsequently, choosing a positive Lyapunov function candidate $V_{d_\chi} = \frac{1}{2} s_\chi^\top s_\chi$, and substituting (14) into the differentiation of V_{d_χ} , one has

$$\dot{V}_{d_\chi} = s_\chi^\top \dot{s}_\chi \leq -[\lambda_{d_1}, \lambda_{d_2}, \lambda_{d_3}, \lambda_{d_4}] |s_\chi|, \quad (15)$$

which indicates that the sliding surface s_χ converge to zero in finite time, i.e., $s_\chi = \mathbf{0}_{3 \times 1}$. Therefore, one can achieve $\dot{e}_\chi = -c_{\alpha_\chi} e_\chi^{[\alpha_\chi]} - c_{\beta_\chi} e_\chi^{[\beta_\chi]}$ from (10). Based on *Lemma 1*, the equilibrium point of (13) is finite time stable, i.e.,

$$\lim_{t \rightarrow t_{f_\chi}} e_\chi = \mathbf{0}_{3 \times 1}, \quad (16)$$

wherein t_{f_χ} represents the finite convergence time. Define the estimation error of the disturbance U_χ as $\tilde{D}_\chi = U_\chi - \hat{D}_\chi$, from (11) and (13)-(16), it is obvious that $\tilde{D}_\chi$ is bounded and

$$\lim_{t \rightarrow t_{f_\chi}} \tilde{D}_\chi = \mathbf{0}_{3 \times 1}. \quad (17)$$

Thus, *Theorem 1* is proved.

B. Controller Development

To achieve the control objective, define the error signal $e = [e_{\xi_1}^\top, e_{\xi_2}^\top]^\top = [e_1, e_2, e_3, e_4]^\top \in \mathbb{R}^4$. Through the energy analysis of the collaborative aerial transportation system, the following energy storage function can be constructed:

$$V = \frac{1}{2} \dot{q}^\top M \dot{q} + \frac{1}{2} mgl_1 (1 - C_1) + \frac{1}{2} mgl_2 (1 - C_2) + mgl_3 (1 - C_3) + \sum_{i=1}^4 k_{p_i} [\ln(\cosh e_i)]. \quad (18)$$

wherein $k_{pi} \in \mathbb{R}_+$ denote positive control gain. Based on *Property 1*, the time derivative of $V(t)$ can be calculated as

$$\dot{V} = \dot{\mathbf{q}}^\top (\mathbf{U} + \mathbf{D}) + \sum_{i=1}^4 k_{pi} (\dot{e}_i \tanh e_i). \quad (19)$$

Then, the control input $\mathbf{U}_\chi = [u_1, u_2, u_3, u_4]^\top \in \mathbb{R}^4$ is designed in of the following form:

$$\begin{aligned} \mathbf{u} = & -\hat{\mathbf{D}}_\chi - K_P \tanh(\mathbf{e}) - K_D \tanh(\dot{\mathbf{e}}) \\ & - K_S \tanh(\dot{\theta}_1^2 + \dot{\theta}_2^2 + \dot{\theta}_3^2) \tanh(\dot{\mathbf{e}}), \end{aligned} \quad (20)$$

wherein, $K_P = \text{diag}([k_{p1}, k_{p2}, k_{p3}, k_{p4}])$, $K_D = \text{diag}([k_{d1}, k_{d2}, k_{d3}, k_{d4}])$, $K_S = \text{diag}([k_{s1}, k_{s2}, k_{s3}, k_{s4}]) \in \mathbb{R}_+^{4 \times 4}$ denote positive control gain matrices.

Remark 1: The last term of (20) enhances the state coupling between the quadrotor UAVs and the cargo by introducing the cargo swing information into the feedback control, which can improve the performance of the controller in suppressing and eliminating the cargo swing.

C. Stability Analysis

Theorem 2: The proposed control law (20) and the disturbance observer (11)-(12) can drive the quadrotor UAVs to the desired position while suppressing the cargo swing in the presence of external disturbances, i.e.,

$$\begin{aligned} \lim_{t \rightarrow \infty} [\xi_1^\top, \xi_2^\top, \theta^\top, \dot{\xi}_1^\top, \dot{\xi}_2^\top, \dot{\theta}^\top]^\top = \\ [\xi_{1d}^\top, \xi_{2d}^\top, \mathbf{0}_{3 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{3 \times 1}^\top]^\top. \end{aligned} \quad (21)$$

Proof 2: Substituting (20) into (19) can obtained that:

$$\begin{aligned} \dot{V} = & -K_D \dot{\mathbf{e}}^\top \tanh(\dot{\mathbf{e}}) - K_S \dot{\mathbf{e}}^\top \tanh(\dot{\theta}_1^2 + \dot{\theta}_2^2 + \dot{\theta}_3^2) \tanh(\dot{\mathbf{e}}) \\ & + \dot{\mathbf{e}}^\top \tilde{\mathbf{D}}_\chi. \end{aligned} \quad (22)$$

The analysis is divided into two steps:

Step 1: $t < t_{fx}$. From (22), one can derive that

$$\dot{V} \leq -\|K_D\| \|\dot{\mathbf{e}}\|^2 + \|\dot{\mathbf{e}}\| \|\tilde{\mathbf{D}}_\chi\| = -\|\dot{\mathbf{e}}\| (\|K_D\| \|\dot{\mathbf{e}}\| - \|\tilde{\mathbf{D}}_\chi\|).$$

Based on the above inequality, when $\|\dot{\mathbf{e}}\| \geq \frac{1}{\|K_D\|} \|\tilde{\mathbf{D}}_\chi\|$, it is obvious that $\dot{V} \leq 0$, which means the error dynamics is decreasing progressively. On the flip side, $\|\dot{\mathbf{e}}\| < \frac{1}{\|K_D\|} \|\tilde{\mathbf{D}}_\chi\|$ implies that the translation error is bounded since $\tilde{\mathbf{D}}_\chi$ is bounded. In summary, when $t < t_{fx}$, the translation error dynamics of the system is bounded.

Step 2: $t \geq t_{fx}$. Based on (17), (22) can be reduced to:

$$\begin{aligned} \dot{V} = & -K_S \dot{\mathbf{e}}^\top \tanh(\dot{\theta}_1^2 + \dot{\theta}_2^2 + \dot{\theta}_3^2) \tanh(\dot{\mathbf{e}}) \\ & - K_D \dot{\mathbf{e}}^\top \tanh(\dot{\mathbf{e}}) \leq 0, \end{aligned} \quad (23)$$

which indicates that $V(t) \in \mathcal{L}_\infty$. Based on (18), it is straightforward to make the following conclusions:

$$\mathbf{e}, \dot{\xi}_1, \dot{\xi}_2, \dot{\theta} \in \mathcal{L}_\infty \Rightarrow \dot{\mathbf{e}}, \dot{\xi}_1, \dot{\xi}_2 \in \mathcal{L}_\infty. \quad (24)$$

Subsequently, define set Φ as

$$\Phi = \left\{ (\mathbf{e}, \dot{\mathbf{e}}, \theta, \dot{\theta}) \mid \dot{V}(t) = 0 \right\}. \quad (25)$$

Let Γ be the largest invariant set in Φ , based on (23), the following conclusions can be achieved:

$$\begin{aligned} \dot{e}_1 = \dot{e}_2 = \dot{e}_3 = \dot{e}_4 = 0 \Rightarrow \ddot{e}_1 = \ddot{e}_2 = \ddot{e}_3 = \ddot{e}_4 = 0, \\ e_1 = \beta_1, e_2 = \beta_2, e_3 = \beta_3, e_4 = \beta_4, \end{aligned} \quad (26)$$

wherein, $\beta_1, \beta_2, \beta_3, \beta_4$ denote constants to be determined.

Firstly, prove $e_1(t) = e_2(t) = e_3(t) = e_4(t) = 0$ in the invariant set Γ . Substituting (26) into (20) and combining (2)-(5), one can achieve that:

$$\begin{aligned} \frac{1}{4} m (l_1 C_1 \ddot{\theta}_1 - l_2 C_2 \ddot{\theta}_2 + 2l_3 C_3 \ddot{\theta}_3 - l_1 S_1 \dot{\theta}_1^2 + l_2 S_2 \dot{\theta}_2^2 - \\ 2l_3 S_3 \dot{\theta}_3^2) = -k_{p1} \tanh \beta_1, \end{aligned} \quad (27)$$

$$\begin{aligned} \frac{1}{4} m (l_1 S_1 \ddot{\theta}_1 + l_2 S_2 \ddot{\theta}_2 + 2l_3 S_3 \ddot{\theta}_3 + l_1 C_1 \dot{\theta}_1^2 + l_2 C_2 \dot{\theta}_2^2 + \\ 2l_3 C_3 \dot{\theta}_3^2) = -k_{p2} \tanh \beta_2, \end{aligned} \quad (28)$$

$$\begin{aligned} \frac{1}{4} m (l_1 C_1 \ddot{\theta}_1 - l_2 C_2 \ddot{\theta}_2 + 2l_3 C_3 \ddot{\theta}_3 - l_1 S_1 \dot{\theta}_1^2 + l_2 S_2 \dot{\theta}_2^2 - \\ 2l_3 S_3 \dot{\theta}_3^2) = -k_{p3} \tanh \beta_3, \end{aligned} \quad (29)$$

$$\begin{aligned} \frac{1}{4} m (l_1 S_1 \ddot{\theta}_1 + l_2 S_2 \ddot{\theta}_2 + 2l_3 S_3 \ddot{\theta}_3 + l_1 C_1 \dot{\theta}_1^2 + l_2 C_2 \dot{\theta}_2^2 + \\ 2l_3 C_3 \dot{\theta}_3^2) = -k_{p4} \tanh \beta_4. \end{aligned} \quad (30)$$

Integrating both sides of (27)–(30) with respect to time, the following result can be achieved:

$$\frac{1}{4} m (l_1 C_1 \dot{\theta}_1 - l_2 C_2 \dot{\theta}_2 + 2l_3 C_3 \dot{\theta}_3) = -k_{p1} (\tanh \beta_1) t + \eta_1, \quad (31)$$

$$\frac{1}{4} m (l_1 S_1 \dot{\theta}_1 + l_2 S_2 \dot{\theta}_2 + 2l_3 S_3 \dot{\theta}_3) = -k_{p2} (\tanh \beta_2) t + \eta_2, \quad (32)$$

$$\frac{1}{4} m (l_1 C_1 \dot{\theta}_1 - l_2 C_2 \dot{\theta}_2 + 2l_3 C_3 \dot{\theta}_3) = -k_{p3} (\tanh \beta_3) t + \eta_3, \quad (33)$$

$$\frac{1}{4} m (l_1 S_1 \dot{\theta}_1 + l_2 S_2 \dot{\theta}_2 + 2l_3 S_3 \dot{\theta}_3) = -k_{p4} (\tanh \beta_4) t + \eta_4, \quad (34)$$

wherein, $\eta_1, \eta_2, \eta_3, \eta_4$ denote constants to be determined. Analyzing $\beta_1, \beta_2, \beta_3, \beta_4$, take (31) for an example, if $\beta_1 \neq 0$, $l_1 C_1 \dot{\theta}_1 - l_2 C_2 \dot{\theta}_2 + 2l_3 C_3 \dot{\theta}_3 \rightarrow \infty$, when $t \rightarrow \infty$. Considering that $C_1, C_2, C_3 \in \mathcal{L}_\infty$, it is obvious that this conclusion is contradictory to the conclusion $\dot{\theta} \in L_\infty$ in (24), therefore, the assumption $\beta_1 \neq 0$ cannot hold. A similar approach can be used to analyze $\beta_2, \beta_3, \beta_4$, one can achieve that in the invariant set Γ :

$$\beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \Rightarrow e_1 = e_2 = e_3 = e_4 = 0. \quad (35)$$

Secondly, prove $\theta_1(t) = \theta_2(t) = \theta_3(t) = 0$ in the invariant set Γ . Substituting (26) into (6)–(8), one can obtain that:

$$\begin{aligned} l_1 \ddot{\theta}_1 - l_2 C_{1+2} \ddot{\theta}_2 + 2l_3 C_{1-3} \ddot{\theta}_3 + l_2 S_{1+2} \dot{\theta}_2^2 + 2l_3 S_{1-3} \dot{\theta}_3^2 + \\ 2gS_1 = 0 \end{aligned} \quad (36)$$

$$\begin{aligned} l_1 C_{1+2} \ddot{\theta}_1 - l_2 \ddot{\theta}_2 + 2l_3 C_{2+3} \ddot{\theta}_3 - l_1 S_{1+2} \dot{\theta}_1^2 - 2l_3 S_{2+3} \dot{\theta}_3^2 - \\ 2gS_2 = 0 \end{aligned} \quad (37)$$

$$\begin{aligned} l_1 C_{1-3} \ddot{\theta}_1 - l_2 C_{2+3} \ddot{\theta}_2 + 2l_3 \ddot{\theta}_3 - l_1 S_{1-3} \dot{\theta}_1^2 + l_2 S_{2+3} \dot{\theta}_2^2 + \\ 2gS_3 = 0. \end{aligned} \quad (38)$$

Substituting (35) into (27)-(28) yields

$$l_1 C_1 \ddot{\theta}_1 - l_2 C_2 \ddot{\theta}_2 + 2l_3 C_3 \ddot{\theta}_3 - l_1 S_1 \dot{\theta}_1^2 + l_2 S_2 \dot{\theta}_2^2 - 2l_3 S_3 \dot{\theta}_3^2 = 0, \quad (39)$$

$$l_1 S_1 \ddot{\theta}_1 + l_2 S_2 \ddot{\theta}_2 + 2l_3 S_3 \ddot{\theta}_3 + l_1 C_1 \dot{\theta}_1^2 + l_2 C_2 \dot{\theta}_2^2 + 2l_3 C_3 \dot{\theta}_3^2 = 0. \quad (40)$$

Subsequently, by multiplying (39) and (40) by C_1 and S_1 respectively, and adding them together, the following result can be derived:

$$l_1 \ddot{\theta}_1 - l_2 C_{1+2} \ddot{\theta}_2 + 2l_3 C_{1-3} \ddot{\theta}_3 + l_2 S_{1+2} \dot{\theta}_2^2 + 2l_3 S_{1-3} \dot{\theta}_3^2 = 0. \quad (41)$$

Combining (36) and (41), it can be obtained that

$$S_1 = 0 \Rightarrow \theta_1 = 0, \dot{\theta}_1 = 0. \quad (42)$$

The following conclusions can be drawn from a similar analysis:

$$\theta_2 = 0, \theta_3 = 0, \dot{\theta}_2 = 0, \dot{\theta}_3 = 0. \quad (43)$$

Based on (35), (42), and (43), it is known that the largest invariant set Γ only contains the equilibrium point $[\xi_1^\top, \xi_2^\top, \theta^\top, \dot{\xi}_1^\top, \dot{\xi}_2^\top, \dot{\theta}^\top]^\top = [\xi_{1d}^\top, \xi_{2d}^\top, \mathbf{0}_{3 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{2 \times 1}^\top, \mathbf{0}_{3 \times 1}^\top]^\top$. According to LaSalle's invariance theorem [19], the conclusion in (21) can be drawn.

Above all, *Theorem 2* is proved.

IV. SIMULATION RESULTS

In this section, the basic performance and robustness of the proposed control scheme are verified by two sets of numerical simulations. The PD (proportional differential) controller is chosen as the comparison method to evaluate the proposed control scheme in terms of UAVs positioning and swing suppression.

The physical parameters of the system are chosen as $M_1 = M_2 = 5\text{kg}$, $m = 2\text{kg}$, $l_1 = l_2 = 2\text{m}$, $l_3 = 0.1\text{m}$, $g = 9.8\text{m/s}^2$. The control gains of the proposed method are selected as $K_P = \text{diag}([4, 4, 4, 4])$, $K_D = \text{diag}([8, 8, 8, 8])$, $K_S = \text{diag}([10, 10, 10, 10])$, $c_{\alpha_\chi} = \text{diag}([5, 5, 5, 5])$, $c_{\beta_\chi} = \text{diag}([5, 5, 5, 5])$, $\alpha_\chi = 0.5$, $\beta_\chi = 2$, $\lambda_{d_\chi} + \mu_{d_\chi} = \text{diag}([30, 30, 30, 30])$. The control gains of the PD controller are selected as $K_P = \text{diag}([5, 5, 5, 5])$, $K_D = \text{diag}([14, 14, 14, 14])$.

The initial and the desired position of the quadrotor UAVs are set as $\xi_1(0) = [0, 3]^\top \text{m}$, $\xi_2(0) = [1, 3]^\top \text{m}$, $\xi_{1d} = [2, 5]^\top \text{m}$, $\xi_{2d} = [3, 5]^\top \text{m}$.

A. Basic Performance Verification Simulation

The positions of two UAVs and the cargo angles are shown in Fig. 2, from which one can see that compared to the PD controller, the proposed method can achieve similar UAV positioning performance and has better performance in terms of cargo swing suppression.

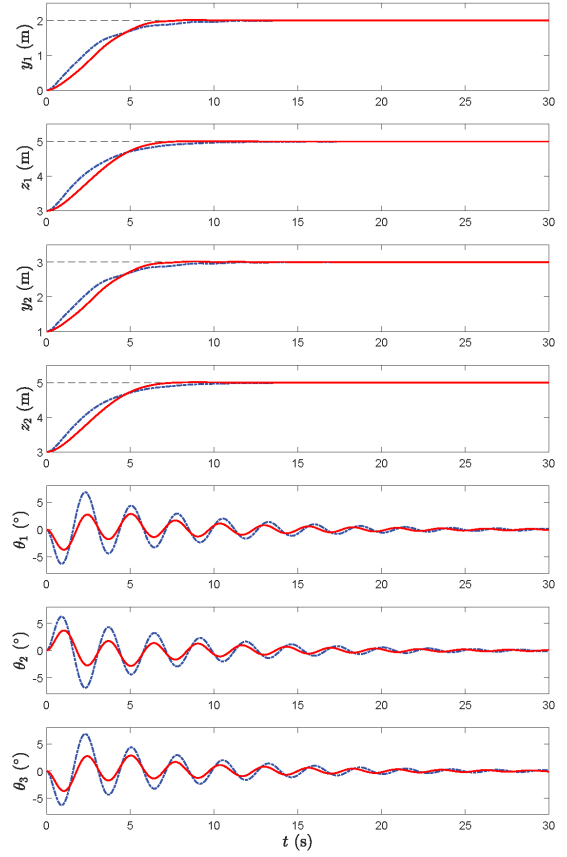


Fig. 2. Result for Basic Performance Verification Simulation. (Red solid line: proposed method. Blue dotted line: PD controller. Black dotted line: desired position.)

B. Robustness Verification Simulation

To simulate the external disturbance, a disturbance with an amplitude of $4N$ is applied to each direction of the two quadrotor UAVs at $20 - 25\text{s}$.

The simulation results are shown in Fig. 3 and Fig. 4. Specifically, Fig. 3 shows the positions of two UAVs and the load angles, and Fig. 4 shows the estimated value of the disturbance given by the observer. It can be seen that the proposed method has superior performance in resisting disturbance. The simulation verifies that the proposed controller has excellent robustness.

V. CONCLUSION

This paper focuses on the robust controller design of the dual-UAV collaborative aerial transportation system. In particular, by rewriting the form of the system dynamic model, the finite time disturbance observer can be utilized to suppress external disturbances. Subsequently, an effective control scheme is designed based on energy analysis, which can accurately and quickly drive UAVs to the desired position, and effectively eliminate cargo swing. Then, Lyapunov techniques and LaSalle's Invariance Theorem are utilized to prove the stability of the closed-loop system. Simulation results show the superior control performance and robustness of the

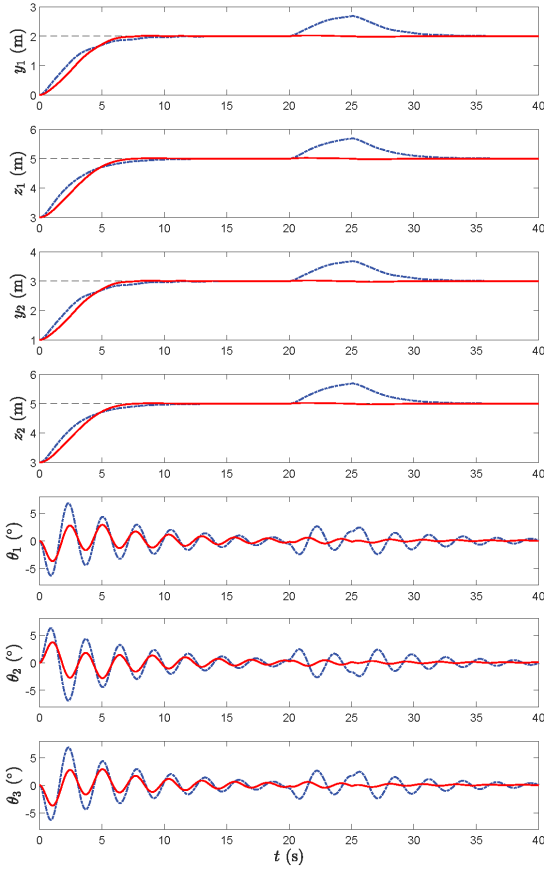


Fig. 3. Result for Robustness Verification Simulation. (Red solid line: proposed method. Blue dotted line: PD controller. Black dotted line: desired position.)

proposed controller. In the ensuing works, the control scheme proposed in this paper will be further extended and applied to the actual cooperative aerial transportation system containing more UAVs.

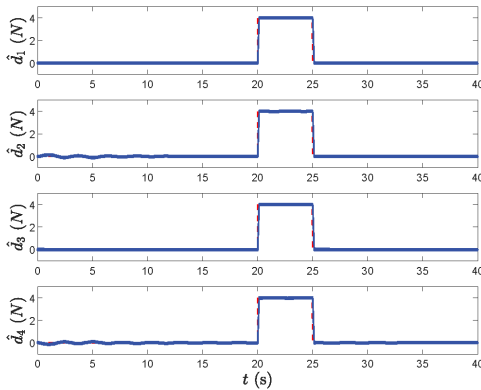


Fig. 4. Result of Observer. (Blue solid line: observed value. Red dotted line: real value.)

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